Trial i	Position x_i, cm	Temperature T_i, °C	x_i^2	x_iT_i	T_i^2	$a + bx_i$	Δ_i^2
1	1.0	15.6	1.0	15.6	243.36	14.2	1.96
2	2.0	17.5	4.0	35.0	306.25	23.6	37.21
3	3.0	36.6	9.0	109.8	1339.56	33.0	12.96
4	4.0	43.8	16.0	175.2	1918.44	42.4	1.96
5	5.0	58.2	25.0	291.0	3387.24	51.8	40.96
6	6.0	61.6	36.0	369.6	3794.56	61.2	0.16
7	7.0	64.2	49.0	449.4	4121.64	70.6	40.96
8	8.0	70.4	64.0	563.2	4956.16	80.0	92.16
9	9.0	98.8	81.0	889.2	9761.44	89.4	88.36
SUM	45.0	466.7	285.0	2898.0	29,828.65		316.69

$$\Delta = N\Sigma x_i^2 - (\Sigma x_i)^2 = 9(285) - (45)^2 = 540$$

$$a = \frac{\Sigma x_i^2 \Sigma T_i - \Sigma x_i \Sigma x_i T_i}{\Delta} = \frac{285(466.7) - 45(2898)}{540} = 4.8$$

$$b = \frac{N\Sigma x_i T_i - \Sigma x_i \Sigma T_i}{\Delta} = \frac{9(2898) - 45(466.7)}{540} = 9.4$$

$$s^2 = \frac{1}{N-2}\Sigma(T_i - a - bx_i)^2 = \frac{1}{N-2}\Sigma(\Delta_i)^2 = \tfrac{1}{7}(316.69) = 45.24$$

$$s^2 = \frac{1}{N-2}[\Sigma T_i^2 + Na^2 + b^2\Sigma x_i^2 - 2(a\Sigma T_i - ab\Sigma x_i + b\Sigma x_i T_i)]$$

$$= \tfrac{1}{7}[29{,}828.65 + 207.36 + 25{,}182.60 - 2(2240.16 - 2030.40 + 27{,}241.2)] = 45.24$$

$$\sigma_a^2 \simeq \frac{s^2\Sigma x_i^2}{\Delta} = 23.9 \qquad \sigma_a \simeq 4.9°\text{C}$$

$$\sigma_b^2 \simeq \frac{Ns^2}{\Delta} = 0.754 \qquad \sigma_b \simeq 0.87°\text{C/cm}$$

$$\sigma \simeq \sqrt{s^2} = 6.7 \qquad \text{P.E.} = \tfrac{2}{3}\sigma \simeq 4.5°\text{C}$$